MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **WEEKLY CHAPTER TEST – 06 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 6: Triangles** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, the length of her shadow after 4 seconds is: [1]

(a) 1.2 m (b) 1.6 m (c) 2.0 m (d) 2.4 m

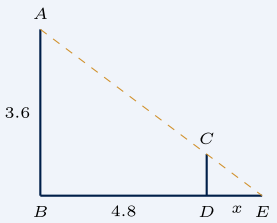
▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 1.6 m

Given: Lamp height $AB = 3.6$ m; girl height $CD = 90$ cm $= 0.9$ m; speed $= 1.2$ m/s; time $= 4$ s.

Distance walked in 4 s: $BD = 1.2 \times 4 = 4.8$ m. Let shadow $DE = x$ m.

In $\triangle ABE$ and $\triangle CDE$, $\angle B = \angle D = 90°$ and $\angle E$ is common, so $\triangle ABE \sim \triangle CDE$ (AA):



Formula / Theorem Used: $\dfrac{AB}{CD} = \dfrac{BE}{DE} = \dfrac{BD + DE}{DE}$

$$\frac{3.6}{0.9} = \frac{4.8 + x}{x} \implies 4 = \frac{4.8 + x}{x}$$

$$4x = 4.8 + x \implies 3x = 4.8 \implies \boxed{x = \mathbf{1.6}\ \text{m}}$$

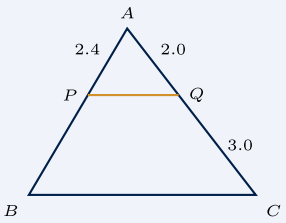
Q2. In $\triangle ABC$, P and Q are points on AB and AC respectively such that $PQ \parallel BC$. If $AP = 2.4$ cm, $AQ = 2.0$ cm, and $QC = 3.0$ cm, then the length of AB is: [1]

(a) 4.8 cm (b) 6.0 cm (c) 5.0 cm (d) 7.2 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 6.0 cm

Given: In $\triangle ABC$, $PQ \parallel BC$, $AP = 2.4$ cm, $AQ = 2.0$ cm, and $QC = 3.0$ cm.



We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: Since $PQ \parallel BC$, $\dfrac{AP}{PB} = \dfrac{AQ}{QC}$

Substituting the given lengths:

$$\frac{2.4}{PB} = \frac{2.0}{3.0} \implies PB = \frac{2.4 \times 3}{2} = 3.6 \text{ cm}$$

$$AB = AP + PB = 2.4 + 3.6 = \boxed{\mathbf{6.0}\text{ cm}}$$

Q3. Which of the following is NOT a valid criterion for similarity of two triangles? [1]

(a) SAS (b) AAA (c) SSA (d) SSS

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) SSA

We know the valid similarity criteria for triangles (NCERT Class X):

Formula / Theorem Used: Valid similarity criteria are **AAA (or AA)**, **SSS**, and **SAS** (where the equal angle must lie *between* the two proportional sides).

In **SSA**, the equal angle is *not included* between the proportional sides, which can yield two non-similar triangles (ambiguous case). Therefore, **SSA** is not a valid similarity criterion.

Q4. If in triangles ABC and PQR, $\angle A = \angle Q = 40°$, $AB : QR = AC : QP$, and $\angle R = 65°$, then the degree measure of $\angle B$ is: [1]

(a) 65° (b) 75° (c) 40° (d) 85°

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 65°

Given: In $\triangle ABC$ and $\triangle QRP$, $\dfrac{AB}{QR} = \dfrac{AC}{QP}$ and their included angles satisfy $\angle A = \angle Q = 40°$.

We know the SAS Similarity Criterion:

Formula / Theorem Used: If two sides of one triangle are proportional to two sides of another triangle and the included angles are equal, the triangles are similar.

Applying SAS similarity carefully with exact vertex correspondence ($A \leftrightarrow Q$, $B \leftrightarrow R$, $C \leftrightarrow P$):

$$\triangle ABC \sim \triangle QRP \implies \angle B = \angle R = \boxed{\mathbf{65°}}$$

Q5. Assertion (A): If in $\triangle ABC$ and $\triangle PQR$, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, then $\triangle ABC \sim \triangle PQR$.

Reason (R): If corresponding sides of two triangles are proportional, then their corresponding angles are equal and the triangles are similar. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (d) Assertion (A) is false but Reason (R) is true.

We know the SSS Similarity Criterion (Theorem 6.4):

Formula / Theorem Used: If sides of two triangles are proportional, equal angles lie opposite proportional sides, fixing vertex correspondence.

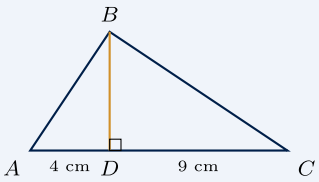
Evaluating (A) & (R): Since $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, vertex $A \leftrightarrow Q$, $B \leftrightarrow R$, and $C \leftrightarrow P$. Therefore, $\triangle ABC \sim \triangle QRP$ (and *not* $\triangle PQR$ unless the triangles are equilateral). Hence, **Assertion (A) is false**, whereas **Reason (R) is true**.

SECTION B: Descriptive Questions MATHS MANTHAN | $(1 \times 2 + 3 \times 3 = 11$ Marks)

Q6. In $\triangle ABC$, $\angle B = 90°$ and $BD \perp AC$. If $AD = 4$ cm and $CD = 9$ cm, find the length of altitude BD. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: In right $\triangle ABC$, $\angle ABC = 90°$, $BD \perp AC$, $AD = 4$ cm, $CD = 9$ cm.



We know the Right-Triangle Altitude Similarity Theorem:

Formula / Theorem Used: If a perpendicular is drawn from the right-angle vertex to the hypotenuse, triangles on both sides are similar to each other.

Step 1: Proving $\triangle ADB \sim \triangle BDC$ [1 Mark]

In right $\triangle ADB$, $\angle ADB = 90°$, so $\angle A + \angle ABD = 90° \implies \angle ABD = 90° - \angle A$.

Also, $\angle ABC = 90° \implies \angle ABD + \angle CBD = 90° \implies \angle CBD = 90° - \angle ABD = \angle A$.

In $\triangle ADB$ and $\triangle BDC$: $\angle A = \angle CBD$ (proved) and $\angle ADB = \angle BDC = 90°$. Hence, by AA similarity, $\triangle ADB \sim \triangle BDC$.

Step 2: Computing Altitude BD [1 Mark]

Corresponding sides of similar triangles $\triangle ADB$ and $\triangle BDC$ are proportional:

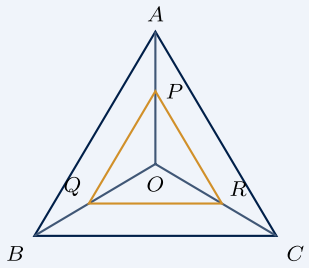
$$\frac{AD}{BD} = \frac{BD}{CD} \implies BD^2 = AD \times CD = 4 \times 9 = 36$$

Taking positive square root:

$$\boxed{BD = \sqrt{36} = \mathbf{6} \text{ cm}}$$

Q7. In the given figure, O is a point inside $\triangle ABC$. Points P, Q, R are taken on OA, OB, OC such that $OP = 2$ cm, $OQ = 3$ cm, $OR = 4$ cm and $PQ \parallel AB$, $QR \parallel BC$. Prove that $PR \parallel AC$ and $\triangle PQR \sim \triangle ABC$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: Point O inside $\triangle ABC$; P, Q, R on OA, OB, OC respectively ($OP = 2, OQ = 3, OR = 4$ cm) such that $PQ \parallel AB$ and $QR \parallel BC$.
To Prove: (i) $PR \parallel AC$, and (ii) $\triangle PQR \sim \triangle ABC$.

We know the Basic Proportionality Theorem (Thales' Theorem) and its Converse:

Formula / Theorem Used: BPT gives proportional segments from parallel lines; Converse of BPT proves parallelism from equal ratios.

Step 1: Applying BPT in $\triangle OAB$ and $\triangle OBC$ [1 Mark]

In $\triangle OAB$, since $PQ \parallel AB$, by BPT (and its corollary for full sides):

$$\frac{OP}{PA} = \frac{OQ}{QB} \quad \text{— (i)} \qquad \text{and} \qquad \frac{OP}{OA} = \frac{OQ}{OB} \quad \text{— (ii)}$$

In $\triangle OBC$, since $QR \parallel BC$, by BPT (and its corollary for full sides):

$$\frac{OQ}{QB} = \frac{OR}{RC} \quad \text{— (iii)} \qquad \text{and} \qquad \frac{OQ}{OB} = \frac{OR}{OC} \quad \text{— (iv)}$$

Step 2: Proving $PR \parallel AC$ [1 Mark]

Equating (i) and (iii) in $\triangle OAC$: $\dfrac{OP}{PA} = \dfrac{OR}{RC}$.
Since PR divides OA and OC in the same ratio, by the **Converse of BPT**: **$PR \parallel AC$**.

Step 3: Proving $\triangle PQR \sim \triangle ABC$ [1 Mark]

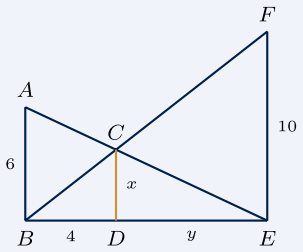
By AA similarity (since $PQ \parallel AB, QR \parallel BC, PR \parallel AC$):
$\triangle OPQ \sim \triangle OAB \implies \dfrac{PQ}{AB} = \dfrac{OQ}{OB}$; $\triangle OQR \sim \triangle OBC \implies \dfrac{QR}{BC} = \dfrac{OQ}{OB}$; $\triangle OPR \sim \triangle OAC \implies \dfrac{PR}{AC} = \dfrac{OP}{OA}$.
Combining with (ii), $\dfrac{PQ}{AB} = \dfrac{QR}{BC} = \dfrac{PR}{AC}$. By **SSS similarity criterion**:

$$\therefore PR \parallel AC \text{ and } \triangle PQR \sim \triangle ABC \quad \text{[Hence Proved]}$$

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Q8. In the given figure, $AB \parallel CD \parallel EF$. If $AB = 6$ cm, $CD = x$ cm, $EF = 10$ cm, $BD = 4$ cm, and $DE = y$ cm, calculate the values of x and y. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: Parallel segments $AB \parallel CD \parallel EF$ standing on baseline BDE, with $AB = 6$ cm, $CD = x$ cm, $EF = 10$ cm, $BD = 4$ cm, $DE = y$ cm, so $BE = y + 4$.
We know the AA Similarity Criterion:

Formula / Theorem Used: If two pairs of corresponding angles of two triangles are equal, their corresponding sides are proportional.

Step 1: Setting up Ratios from Similar Triangles [1 Mark]
1. In $\triangle EDC$ and $\triangle EBA$, since $CD \parallel AB$, $\angle EDC = \angle EBA$ (corresponding) and $\angle E$ is common $\implies \triangle EDC \sim \triangle EBA$ (AA):

$$\frac{CD}{AB} = \frac{DE}{BE} \implies \frac{x}{6} = \frac{y}{y+4} \quad \text{— (i)}$$

2. In $\triangle BDC$ and $\triangle BEF$, since $CD \parallel EF$, $\angle BDC = \angle BEF$ (corresponding) and $\angle B$ is common $\implies \triangle BDC \sim \triangle BEF$ (AA):

$$\frac{CD}{EF} = \frac{BD}{BE} \implies \frac{x}{10} = \frac{4}{y+4} \quad \text{— (ii)}$$

Step 2: Solving for x [1 Mark]
Adding equations (i) and (ii):

$$\frac{x}{6} + \frac{x}{10} = \frac{y}{y+4} + \frac{4}{y+4} = \frac{y+4}{y+4} = 1$$

$$\frac{5x+3x}{30} = 1 \implies \frac{8x}{30} = 1 \implies x = \frac{30}{8} = \frac{15}{4} = 3.75 \text{ cm}$$

Step 3: Solving for y [1 Mark]
Substituting $x = \frac{15}{4}$ into equation (ii):

$$\frac{15/4}{10} = \frac{4}{y+4} \implies \frac{3}{8} = \frac{4}{y+4}$$

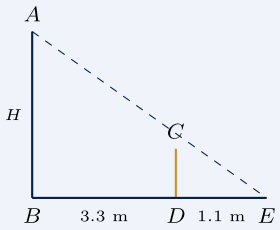
$$3(y+4) = 32 \implies 3y = 20 \implies y = \frac{20}{3} \text{ cm}$$

$$\boxed{x = \mathbf{3.75} \text{ cm } \left(\frac{15}{4} \text{ cm}\right), \qquad y = \frac{\mathbf{20}}{\mathbf{3}} \text{ cm}}$$

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Q9. A person 1.65 m tall stands 3.3 m away from a lamppost and observes his shadow to be 1.1 m long on the ground. Find the height of the lamppost. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Height of person $CD = 1.65$ m; distance from lamppost $BD = 3.3$ m; shadow length $DE = 1.1$ m. Let lamppost height be $AB = H$ m. Total base $BE = BD + DE = 3.3 + 1.1 = 4.4$ m. We know the AA Similarity Criterion:

Formula / Theorem Used: If two angles of one triangle equal two angles of another triangle, the triangles are similar and their sides are proportional.

Step 1: Proving $\triangle ABE \sim \triangle CDE$ [1.5 Marks]

In $\triangle ABE$ and $\triangle CDE$:

1. $\angle ABE = \angle CDE = 90^\circ$ (Lamppost and person stand vertically on ground)
2. $\angle AEB = \angle CED$ (Common angle of elevation of the light ray)

By the **AA similarity criterion**, $\triangle ABE \sim \triangle CDE$.

Step 2: Finding Lamppost Height AB [1.5 Marks]

Corresponding sides of similar triangles are proportional:

$$\frac{AB}{CD} = \frac{BE}{DE} \implies \frac{H}{1.65} = \frac{4.4}{1.1} = 4$$

$$H = 4 \times 1.65 = 6.6 \text{ m} \implies \boxed{\text{Height of lamppost } = \mathbf{6.6} \text{ m}}$$

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Q10. Case Study 06: Night-Time Streetlight & Moving Pedestrian Kinematics

A municipal street light is mounted at the top of a pole 3.6 m above the pavement. A student of height 90 cm (0.9 m) walks away from the base of the lamppost along a straight sidewalk at a steady speed of 1.2 m/s. [4]

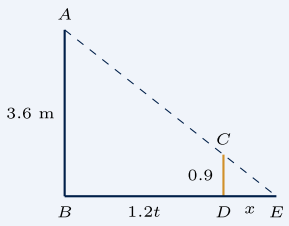
(i) **[1 Mark]** How far from the base of the lamppost is the student after 4 seconds?

(ii) **[1 Mark]** Find the length of the student's shadow after 4 seconds.

(iii) **[2 Marks]** At what rate is the tip of his shadow moving away from the base of the lamppost?

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Lamp height $AB = 3.6$ m; student height $CD = 0.9$ m; walking speed $v = 1.2$ m/s. Let B be the lamppost base, D the student's position, and E the tip of his shadow.
We know the kinematics relation and AA similarity property:



Formula / Theorem Used: Distance = Speed × Time and $\dfrac{AB}{CD} = \dfrac{BE}{DE}$

Sub-part (i): Distance from the Base of the Lamppost after 4 seconds [1 Mark]

$$BD = \text{Speed} \times \text{Time} = 1.2 \text{ m/s} \times 4 \text{ s} = 4.8 \text{ m} \implies \boxed{\text{Distance } BD = \mathbf{4.8} \text{ m}}$$

Sub-part (ii): Length of Student's Shadow after 4 seconds [1 Mark]

In $\triangle ABE$ and $\triangle CDE$, $\angle B = \angle D = 90^\circ$ and $\angle E$ is common $\implies \triangle ABE \sim \triangle CDE$ (AA similarity).
Let $DE = x$ m at $t = 4$ s. Then $BE = BD + DE = 4.8 + x$:

$$\frac{AB}{CD} = \frac{BE}{DE} \implies \frac{3.6}{0.9} = \frac{4.8+x}{x} \implies 4 = \frac{4.8+x}{x} \implies 4x = 4.8+x \implies 3x = 4.8 \implies \boxed{x = \mathbf{1.6} \text{ m}}$$

Sub-part (iii): Rate of Movement of Shadow Tip E [2 Marks]

At any time t seconds, the student's distance is $BD = 1.2t$ m. Let his shadow length at time t be $DE = x$ m.
From $\triangle ABE \sim \triangle CDE$:

$$\frac{3.6}{0.9} = \frac{1.2t + x}{x} \implies 4x = 1.2t + x \implies 3x = 1.2t \implies x = 0.4t \text{ m}$$

Therefore, the total distance of the shadow tip E from lamppost base B at any time t is:

$$BE = BD + DE = 1.2t + 0.4t = 1.6t \text{ m}$$

Since distance BE increases uniformly at 1.6 m every second (uniform speed $= \dfrac{BE}{t} = 1.6$ m/s):

$$\boxed{\text{Rate of movement of shadow tip } = \mathbf{1.6} \text{ m/s}}$$

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